

Regional Sales Performance Analysis

1. Project Overview

This project analyzes the branches (regions, states, and cities) of a major supermarket chain that have recorded poor sales performance, based on sales, order, and customer data from across the United States, using transaction data from 9,800 transactions covering various product categories

The results of this analysis will serve as the basis for determining priorities for investigation and operational improvements.

2. Dataset Summary

- **Rows:** 9,800 (Raw Data) → 9,799 (After Removed Duplicate Row)

- **Columns:** 18

- **Key Features:**

- a. Order Details**

- (Order ID, Order Date, Ship Date, Ship Mode)

- b. Customer Details**

- (Customer ID, Customer Name, Segment, Country, City, State, Postal Code, Region)

- c. Product Details**

- (Product ID, Category, Sub-Category, Product Name, Sales)

- **Missing Data:** 11 values in Postal Code column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics in both numeric and categorical data.

	Row ID	Postal Code	Sales
count	9800.000000	9789.000000	9800.000000
mean	4900.500000	55273.322403	230.769059
std	2829.160653	32041.223413	626.651875
min	1.000000	1040.000000	0.444000
25%	2450.750000	23223.000000	17.248000
50%	4900.500000	58103.000000	54.490000
75%	7350.250000	90008.000000	210.605000
max	9800.000000	99301.000000	22638.480000

	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Region	Product ID	Category	Sub-Category	Product Name
count	9800	9800	9800	9800	9800	9800	9800	9800	9800	9800	9800	9800	9800	9800	9800
unique	4922	1230	1326	4	793	793	3	1	529	49	4	1861	3	17	1849
top	CA-2018-100111	05/09/2017	26/09/2018	Standard Class	WB-21850	William Brown	Consumer	United States	New York City	California	West	OFF-PA-10001970	Office Supplies	Binders	Staple envelope
freq	14	38	34	5859	35	35	5101	9800	891	1946	3140	19	5909	1492	47

- **Missing Data Handling:** Checked for null values and imputed missing values in the Postal Code column with 5041 to represent the Vermont state.
- **Duplicate Data Handling:** Checked for duplicated values (except for the **Row ID**) and removed them by remaining only one row.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analytics: From SQL to Power BI Visualization.

We performed structured analysis in PostgreSQL by using drill-down analysis, not showing all region-state-city combinations in a single query. That approach would result in a very long and hard-to-read table.

Analysis Approach

Instead, each layer is used to filter the next layer:

- **Layer 1 — Regional**

Compare *total_sales*, *AOV*, *Sales per Customer (SPC)*, and *Orders per Customer (OPC)* across regions. Identify regions with Overall *Critical* / *Warning* KPIs.

- **Layer 2 — State**

Drill down into the states within the region with problems (South) and compare them to the national benchmark (AOV) and regional benchmarks (SPC, OPC).

- **Layer 3 — City**

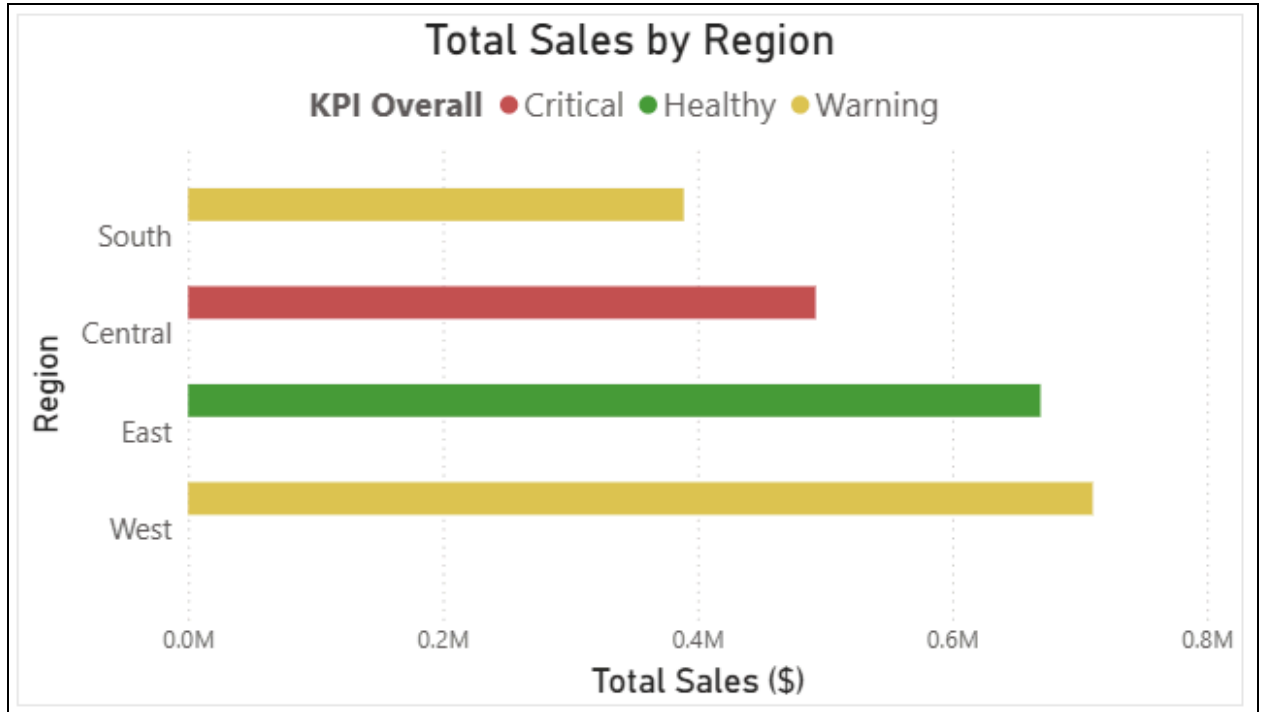
Drill down to the city within the priority states (*Florida*, *North Carolina*, and *Tennessee*) to identify the specific locations with the poorest performance.

KPI Definition

KPI	Definition	Benchmark
AOV (Average Order Value)	Total Sales / Number of Orders	National AOV (\$459.42)
SPC (Average Order Value)	Total Sales / Number of Customers	Related regional/state averages
OPC (Average Order Value)	Number of Orders / Number of Customers	Related regional/state averages
KPI Overall	- <i>Healthy</i> (All KPIs fulfill benchmarks) - <i>Warning</i> (Some) - <i>Critical</i> (All belows benchmarks)	Combination of AOV , SPC , and OPC

A. Layer 1 — Region Analysis

Region	Total Sales (\$)	Number of Orders	Number of Customers	AOV (\$)	SPC (\$)	OPC	Gap AOV (\$)	KPI Overall
South	389,151.46	810	509	480.43	764.54	1.59	21.01	Warning
Central	492,646.91	1,156	626	426.17	786.98	1.85	-33.25	Critical
East	669,237.35	1,369	669	488.85	1,000.35	2.05	29.43	Healthy
West	710,219.68	1,587	681	447.52	1,042.91	2.33	-11.9	Warning



Insight

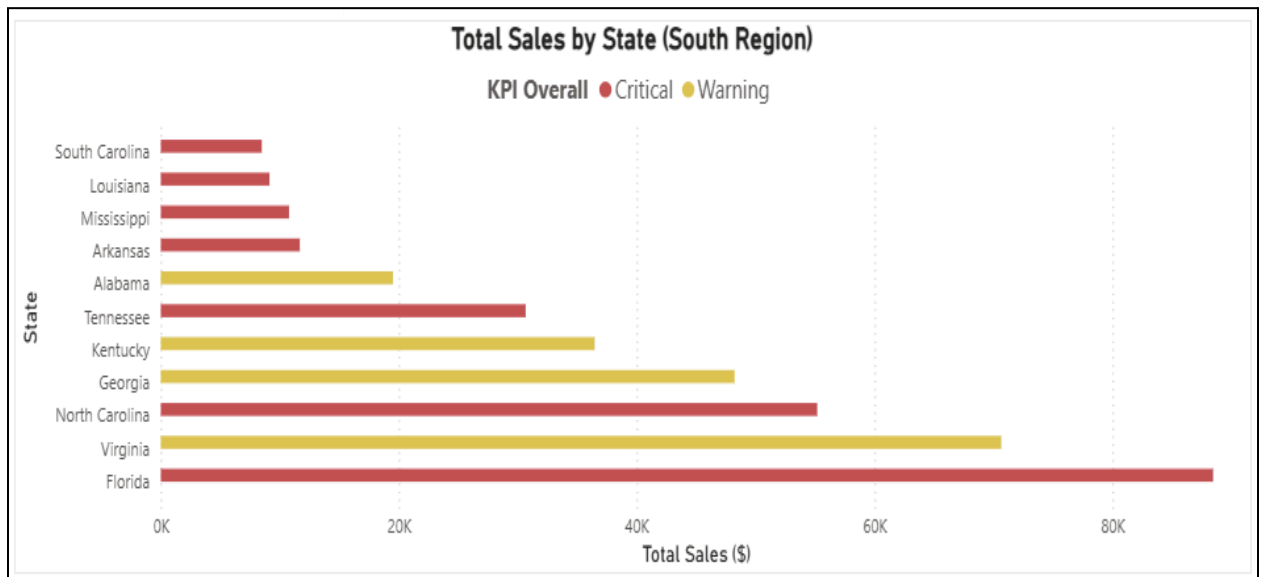
- **South** has the lowest total sales (\$389,151) and an *AOV* above the national benchmark, but its *sales per customer* and *orders per customer* are below the regional average → *Overall KPI: Warning*.
- **Central** region actually had the worst *Overall KPI (Critical)*, with *all three metrics (AOV, SPC, OPC)* falling below the benchmark, even though its *Total Sales* were higher than the South (\$492,646).
- **East** region is the only one rated “*Healthy*”. All three metrics exceed the benchmark, making it a benchmark for best practices (internal benchmark).
- **West** has the highest *total sales*, but its *AOV* is below the national benchmark. Its strong performance is driven by order volume, not the transaction value per order.

Drill-Down Decision

The **South** region was selected for a drill-down to Layer 2. Although the **Central** region has the highest Overall *Critical* KPI, this business issue focuses on the lowest *Total Sales* as the starting point for investigating branches. The **South** region is the most obvious candidate for further geographical analysis.

B. Layer 2 — State Analysis

State	Total Sales (\$)	Number of Orders	Number of Customers	AOV (\$)	SPC (\$)	OPC	Gap AOV (\$)	KPI Overall
South Carolina	8,481.71	20	19	424.09	446.41	1.05	-35.33	Critical
Louisiana	9,131.05	20	20	456.55	456.55	1	-2.87	Critical
Mississippi	10,771.34	26	25	414.28	430.85	1.04	-45.14	Critical
Arkansas	11,678.13	27	27	432.52	432.52	1	-26.9	Critical
Alabama	19,510.64	34	34	573.84	573.84	1	114.42	Warning
Tennessee	30,661.87	91	84	336.94	365.02	1.08	-122.48	Critical
Kentucky	36,458.39	59	56	617.94	651.04	1.05	158.52	Warning
Georgia	48,219.11	87	80	554.24	602.74	1.09	94.82	Warning
North Carolina	55,165.96	134	120	411.69	459.72	1.12	-47.73	Critical
Virginia	70,636.72	115	107	614.23	660.16	1.07	154.81	Warning
Florida	88,436.53	197	178	448.92	496.83	1.11	-10.5	Critical



Insight

- There are **11 states in the South region**, which 6 states are at the *Critical* level, 4 at the *Warning* level, and 0 at the *Healthy* level. This confirms that the issues in the South region are widespread and not limited to just one or two states.
- **Tennessee** has the lowest AOV in the entire **South** region (**\$336.94**) and the largest gap difference (- **\$122.48**) making it a prime candidate for investigation by Layer 3.
- **Florida** contributes 22.7% of total sales in the Southern Region (**\$88,436** out of **\$389,151**) yet remains in the “*Critical*” category. High risk due to its significant contribution to the region as a whole.
- **North Carolina** has the second-highest number of orders (**134**) in the Critical group, but its *AOV* is low (**\$411.69**). This is a large potential for improvement because of the high number of orders.
- **Louisiana** and **Arkansas** are the states at the *Critical* level with “*Orders per Customer*” metric at **1.00**, indicating almost no repeat purchases—a potential customer retention issue.

Drill-Down Decision

Three states were selected for a drill-down to Layer 3:

1. **Tennessee** - The largest *AOV* gap.
2. **North Carolina** - High-volume orders with a significant gap.
3. **Florida** - The largest revenue contributor to the South, but still critical.

C. Layer 3 — City Analysis

State

Florida

North Carolina

Tennessee

Layer 3 – City Analysis

KPI Overall

Critical

Healthy

Warning

City	Total Sales	Number of Orders	Number of Customers	Average Order Value	Sales per Customer	Gap AOV	KPI Overall
Bartlett	88.96	1	1	88.96	88.96	-370.46	Critical
Hendersonville	319.54	2	2	159.77	159.77	-299.65	Critical
Murfreesboro	337.33	4	4	84.33	84.33	-375.09	Critical
Bristol	436.60	4	4	109.15	109.15	-350.27	Critical
Lebanon	488.90	3	3	162.97	162.97	-296.45	Critical
Johnson City	765.85	3	3	255.28	255.28	-204.14	Critical
Smyrna	848.58	4	4	212.14	212.14	-247.28	Critical
Franklin	948.23	6	6	158.04	158.04	-301.38	Critical
Chattanooga	988.80	6	6	164.80	164.80	-294.62	Critical

Insights per State

1. Tennessee

- All cities have an AOV below the national benchmark (**\$459.42**)—not a single one comes close to the benchmark, indicating a structural problem (affecting the entire state), rather than one isolated to a single branch.
- However, for the *Sales per Customer* metric, **4 out of 14 cities (Jackson, Knoxville, Memphis, and Nashville) are above the state benchmark**—resulting in a “Warning” classification rather than “Critical.” A more precise conclusion: **Tennessee’s AOV is uniformly poor, while its Sales per Customer varies.**

2. North Carolina

- **Jacksonville (NC)** — different from **Jacksonville (Florida)** — is the top priority: *Total sales* are \$5,579.86 with an *AOV* of \$242.60 (gap -\$216.82), the highest *order volume* (23) among critical NC cities.
- **Wilmington** has the largest *AOV* gap (-\$376.85) but a low *order volume* (9), so its priority is lower than **Jacksonville**.

3. Florida

- 26 of 33 cities are at the “Critical” level.
- **Miami** is the top priority: It has the highest total sales among *Critical* cities (**\$8,647.83**) with an *AOV* of only **\$262.06 (gap of -\$197.36)**.

- For the record, **Jacksonville (Florida)** is actually in good shape, with the highest AOV (**\$1,029.82**), indicating that there are best practices there that could potentially be replicated.

5. Conclusion: Business Problem Solution

Based on a multi-level analysis by **Region** → **State** → **City**, here are the branches with the worst performance that requires urgent action:

Priority	Location	Key issues
1	Miami, Florida	High order volumes (33), but the AOV is only \$262.06 (a gap of -\$197.36 from the national benchmark)
2	Jacksonville, North Carolina	Highest order volume among Critical cities in NC state(23), where AOV is \$242.60 (gap -\$216.82).
3	Whole cities in Tennessee	Structural issues - AOV across the entire city is below the national benchmark; this is not restricted to a single location.
4	Hialeah & Hollywood, Florida	Large AOV gap but low <i>order volume</i> - medium priority.
5	Wilmington, North Carolina	The largest AOV gap in NC state (-\$376.85), but the volume order was small (9 orders).

6. Analysis Limitations

- Several customers make purchases in more than one state.** As a result, when calculating *the average sales or the number of orders per customer in each region*, there is a possibility that the same customer will be counted more than once. **This can make the results of comparisons between regions slightly less accurate.**

However, the **Average Order Value (AOV)** metric still stays accurate because each transaction is recorded at only one location. This means the key insights from the analysis are still trustworthy.

- The approach to selecting regions for further analysis changed during the analysis process.** Originally, I selected the region with the most features that had the lowest values among all regions, specifically the lowest total sales. As the analysis progressed, I realized that a combination of the severity of the problem and the number of transactions was a better criterion. If that method had been used from the start, the regions selected for further investigation might have been different.

However, these two limitations do not change the main conclusions of the analysis, but they serve as important lessons for improving the quality of the analysis in future editions.

7. Business Recommendations

a. Investigation of mix category products in Tennessee

Lower AOVs across the entire city indicate **the dominance of low-value products** or **overly aggressive discount strategies** at the state level, rather than at the individual branch level.

b. Audit operations in Miami and Jacksonville (NC)

Both branches have *high order volume* but *low transaction value*; improving the *AOV* at high-volume branches has a proportionally significant impact on sales.

c. Evaluate customer retention programs in Louisiana and Arkansas

Orders per Customer = 1.00 menunjukkan hampir tidak ada repeat purchase.

d. Use Jacksonville (Florida) and the East Region as best-practice case studies

Both are performing well, with *AOVs* well above the benchmark, and could provide insights for improving other branches.

8. Next-steps analysis

a. Analyze product categories (product mix) by city to determine why the *AOV* is low.

b. If higher precision is required for *SPC/OPC*, consider calculating benchmarks based on customer-region/state pairs to avoid double-counting caused by customers who make transactions across state lines.